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Digital Signal Intelligence

A New Information Substrate for Insurance

Abstract:

Digital Signal Intelligence (DSI) defines a new, externally observable information substrate for insurance, enabling deterministic, autonomous risk assessment, exposure estimation, loss propensity inference, and pricing at scale.

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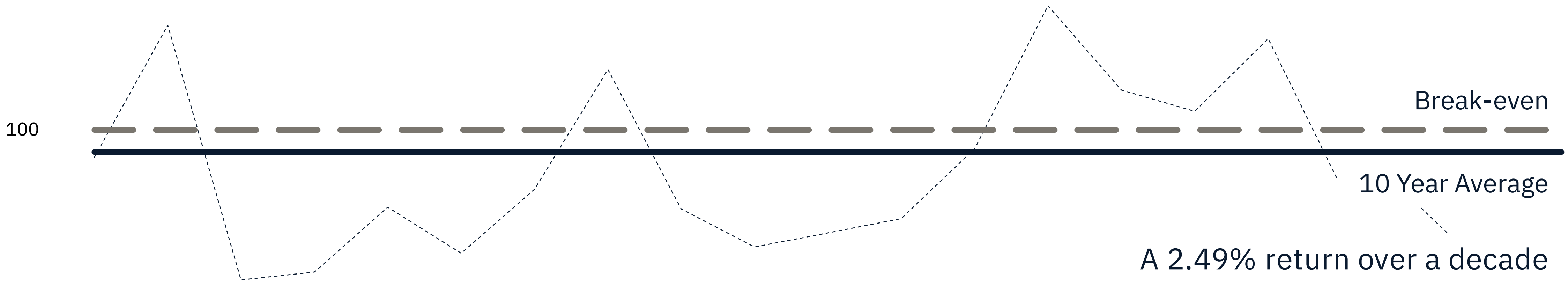
The Automation Myth
Decades of Poor Returns

Twenty years of Lloyd’s combined-ratios show the same truth as every major market: insurance is flat-lining.

Billions spent on process-modernisation and transformation have produced no structural improvement.



Source: [Archived Graphs | III](#) | [Financial results archive - Lloyd’s](#)

The core assumption is that traditional underwriting inputs can be automated.

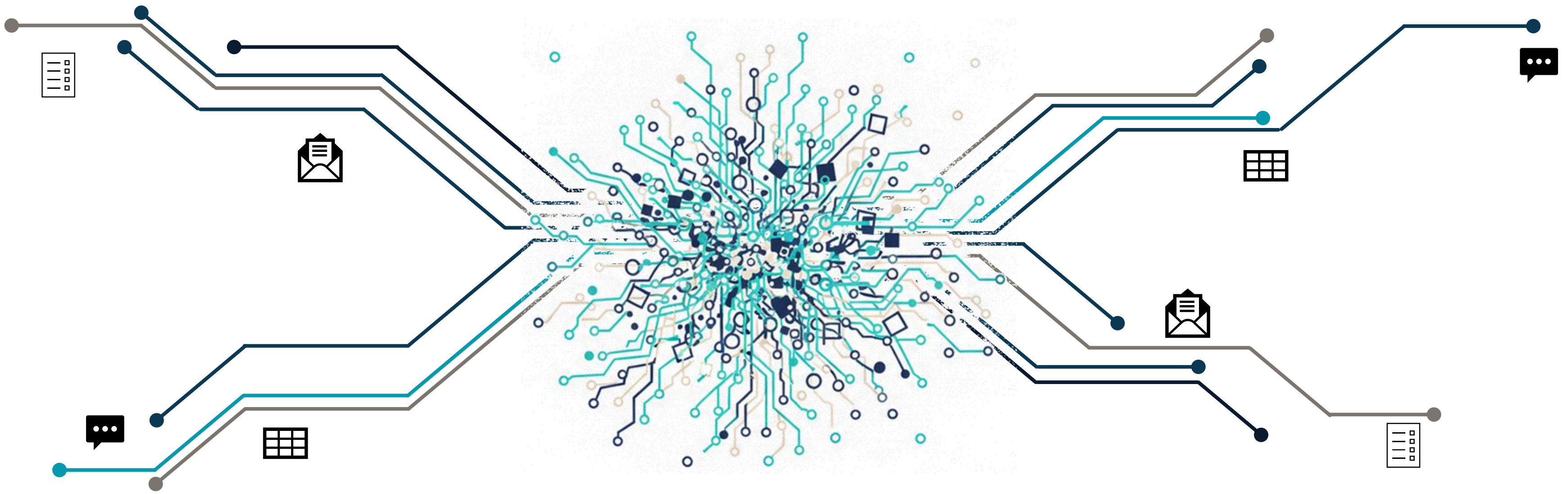
This assumption is incorrect.

The inputs themselves are the constraint.

The Automation Myth

The Information Substrate Problem

Traditional Insurance inputs exist in infinite formats with no canonical schema nor deterministic structure.



No extraction system, including modern or future AI models, can consistently transform these into the inputs required by zero-touch underwriting.

A system built on these inputs cannot achieve deterministic automation.

Autonomous underwriting requires the information substrate to change.

Digital Signal Intelligence
The Core Thesis

Insurance underwriting requires understanding two properties of an organisation:

1. How well it is managed
2. How it behaves over time

Traditional underwriting attempts to infer these properties from self-reported documents, which are subjective, inconsistent, and non-verifiable.

DSI replaces them with observable digital signals.

All organisations operate within a digital ecosystem. Their participation in networks of clients, suppliers, partners, regulators, and certification bodies creates a citation network revealing trust and standing. Their digital trail of operational discipline provides a reliable proxy for maturity. Their hiring activity and investor communications reveal financial performance. Even the absence of expected signals is meaningful because well-run companies are predictable and uniform.

DSI uses these observable signals to create a deterministic, machine-readable foundation for underwriting without requiring self-reported input.

Digital Signal Intelligence

A $1.8 Trillion Global Market — Ready for a New Substrate

DSI applies across the entire insurance landscape because every organisation has a measurable digital footprint.

This creates a universal, cross-coverage opportunity, with seven configured already, and an AI builder to rapidly develop more.

PI
FI
D&O
Cyber
Aerospace
Marine
Energy

DSI is not a product for one line of business — it is a new underwriting substrate for the entire industry.

The Opportunity

The Prize Only Multinationals Can Capture

Every market participant can deploy DSI. Only a multinational group can deploy it everywhere simultaneously — and convert a fragmented collection of regional operations into a single, compounding global risk intelligence network that no single-market competitor can replicate.

The Global Standard

A DSI Tier 2 in Tokyo is the same as a DSI Tier 2 in London. For the first time, the group has a common language for risk quality across every entity it operates — replacing fragmented, jurisdiction-specific underwriting judgment.

The Compounding Moat

Every assessment adds to a proprietary dataset no competitor can access. A risk assessed through the syndicate informs the view of the same counterparty encountered by the US platform. The moat widens automatically with every assessment.

The Existential Imperative

The broker disintermediation threat hits all entities simultaneously. But the group that owns the global standard controls the assessment framework that every submission will eventually be measured against — at any scale, in any market.

Capital Optimisation

A common global tier framework allows the group to demonstrate consistent risk quality distribution to Solvency II, NAIC, MAS, and Lloyd's simultaneously — enabling more precise internal capital allocation and reducing the uncertainty margin.

Digital Signal Intelligence

Why Now?

DSI is not experimental.

It uses the underlying fabric of the modern web as introduced by Google's PageRank in 1998.

1

The Problem

Search engines ranked pages by counting self-reported keywords, resulting in spam.

2

The Breakthrough

PageRank introduced a new information substrate: quality inferred from citation.

3

The Parallel (DSI)

DSI applies the same principle. Risk quality is inferred from observable digital signals.

This fact, coupled with modern cloud and AI technologies provide ideal conditions for success.

The question is if this is feasible. The question is why insurance has taken so long to do so.

Foundational Principles

Ensuring Consistent, Autonomous Processing

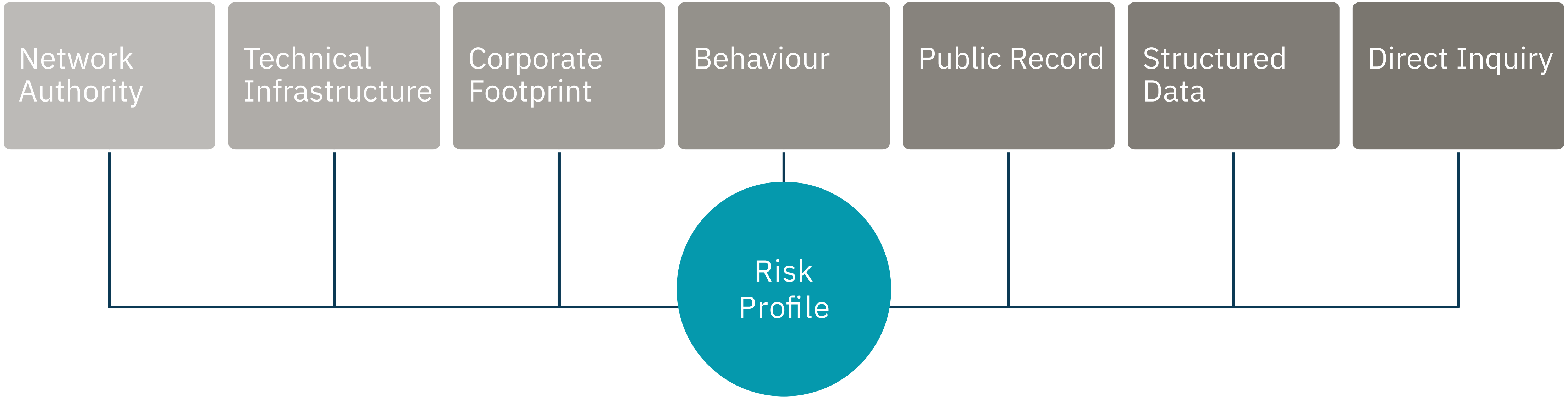
To ensure this, ten mandatory principles define the methodology:

1. External Observability.
   Primary signals must be obtainable without cooperation.
2. Machine Readability.
   All signals must be analysable by automated systems without human interpretation.
3. Network Authority.
   Relationship citation patterns reveal quality
4. Behavioural Inference.
   Observable behaviour is weighted more heavily than stated intentions.
5. Absence as a Signal.
   Absence of expected signals is a measurable indicator of risk.
6. Structured Data Utilisation.
   Authoritative structured data, credit ratings etc, are incorporated directly.
7. Minimal Direct Inquiry.
   Direct questions are optional and constrained to 5-10 per model.
8. Organisational-Level Assessment.
   Risk is assessed at the organisational level; parent behaviour is a proxy for underlying assets.
9. Simplicity in Scoring.
   Models follow a clear scoring flow that is captured and retained at the atomic grain.
10. Agentic Readiness.
    Models must be executable by autonomous systems.

Ensuring consistency, objectivity, and agentic readiness in every assessment.

Signal Architecture
Canonical Categorisation

DSI extracts signals from seven canonical categories that describe how an organisation behaves, operates, and participates in the digital ecosystem.



Signals are externally observable, machine-readable, and verifiable — enabling deterministic scoring without self-reported input.

End-to-End Model Architecture

Structured Workflow

Models convert raw signals into deterministic underwriting outputs through a structured workflow comprising six phases and fourteen discrete steps.

Critically, a post-bind signal loop will transform insurance from a static contract into a dynamic service.

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Three Layer Assessment

Decoupling Complexity for Absolute Clarity

Where traditional underwriting conflates three distinct analytical assessments into a single, subjective, aggregated assessment, DSI separates these into independent, measurable analytical dimensions.

How well does this organisation manage Utilising a PageRank-inspired digital signal citation framework to observe maturity and predictability.	RISK
How large is the insurable A deterministic inference of an organisation's exposure to calculate a size band and complexity score.	EXPOSURE
How frequent and severe is Correlating real-time signal behaviours with historical loss patterns to predict frequency and severity without bias.	LOSS

Each dimension is assessed independently using the same underlying signal infrastructure, then combined into a unified pricing decision.

Worked Example

Assessing "TechFlow"

The following example demonstrates DSI's end-to-end assessment of a mid-market technology company seeking cyber coverage.

It's shown using the same end-to-end architecture previously outlined.

1

Discovery.

- DSI validates techflow.io as the corporate website (95% confidence) and identifies TechFlow Solutions Inc. as the legal entity through corporate registry matching.

2

Model Instantiation.

- Model version created, inputs verified against minimum viable requirements.

3

Signals.

- Parallel extraction including 35 cyber-specific signals.

4

Three Layer Assessment.

- Composite Score: 782/1000 → Tier 2
- Exposure Band: Medium ($10M-$50M TIV)
- Complexity Score: Moderate
- Loss Propensity: Low (score 28 / 100)
- Severity Propensity: Moderate (42 / 100)

5

Pricing.

- Final Premium ($5M limit): $20,250
  Base Rate: 0.45% of Limit
  Loss Modifier: 0.90
- No Other Modifiers Applied.

6

Decision.

- Decision: AUTO-APPROVE
  (Tier 2, confidence 0.87, no triggers)
- Total processing time: 47 seconds.
- Full audit trail persisted.

Competitive Differentiation

Why the Traditional Model Fails Where DSI Wins

DSI occupies a fundamentally different position in the underwriting landscape than existing signal-based tools.

Feature	Traditional	Digital Signal Intelligence
Input	Submissions (PDFs, Excel, Word)	Digital Ecosystem
Process	Manual, OCR, NLP	Deterministic Processing
Integrity	High “Manual Noise" ratio	Zero-Touch, Audit-Persistent
Adjustment	Subjective Overrides	Rule-Based Deterministic Modifiers
Scalability	Linear — bounded by headcount	Logarithmic — bounded by compute
Coverage	Single point-in-time assessment	Continuous monitoring with aware refresh

This is a categorical distinction. DSI is not a better data feed—it is a complete underwriting substrate that happens to consume data feeds, including those from existing vendors, as inputs.

We don't make the old process faster; we make the old process obsolete.

While competitors fight for 10% efficiency gains in data entry, we are rebuilding the foundation.

Economic Impact

Market-level efficiency — distributed across every participant

The insurance market operates with a structurally inefficient information substrate.

DSI does not redistribute this cost. It removes it.

22 pts
Expense ratio improvement

6 – 12 pts
Loss ratio improvement

$130 – 170m
Profit improvement per $500m book

18 – 36 months
Realisation timeframe

Cohort	Metric	Traditional Model	DSI Target
Operational Metrics	Cost per submission	$150–650	< $10
	Straight-through rate	5–15%	60–80%
	Time to quote	3–10 days	< 60 seconds
	Scalability	Linear (headcount)	Logarithmic (compute)

The economic value accrues to whoever deploys first and most completely.

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The Path to Market Standard

Why DSI has the characteristics the traditional model never could

The Traditional Model

300 Years, No Standard

- ✗ Subjective — varies by individual
- ✗ Non-repeatable — inconsistent outcomes
- ✗ Unauditable — judgment not traceable
- ✗ Gameable — based on self-reporting

Digital Signal Intelligence

Built for Standardisation

- ✓ Observable — every signal externally verifiable
- ✓ Repeatable — same entity, same result
- ✓ Auditable — signal → tier → premium chain
- ✓ Non-gameable — drawn from behaviour, not declaration

The same characteristics that made these standards

- FICO Credit Standard
- Google PageRank
- Rating Agency Scores

12–24-month validation → independent evidence base → new capital enters → network effect self-sustains.

Digital Signal Intelligence
The Future of Insurance is Agentic

Digital Signal Intelligence replaces the traditional, self-reported information substrate of insurance with externally observable, machine-readable signals. This enables:

- Deterministic scoring without subjective interpretation
- Autonomous underwriting with deterministic guardrails at scale
- Enhanced Portfolio Steering with Signal weighting optimised at the turn of dial
- Exposure estimation without self-reported schedules
- Loss pattern inference from behavioural signals
- Continuous monitoring and early deterioration detection
- Global scalability without linear headcount growth

Insurers are publicly committing to reducing operational complexity and headcount. However, no traditional underwriting model can achieve this at scale. The bottleneck is the information substrate itself.

DSI is a structural replacement for the information substrate of underwriting, providing a viable path to radical operational efficiency and increased profitability.

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The Opportunity

The Governing Variable for US MGAs

US MGA success is determined by one thing above all others: the quality of the capacity relationship. Capacity providers allocate preferentially to MGAs who demonstrate superior risk selection. Most MGAs cannot prove this beyond experience and lagging loss runs. DSI changes that.

Capacity Proof Package

DSI generates carrier-ready analytical output for every risk decision — composite score breakdown, tier rationale, specific signals flagged, confidence score, and premium derivation. The analytical credential preferred capacity allocation requires.

Adverse Selection Defence

The E&S market attracts risks admitted carriers have declined. DSI's seven signal types surface adverse selection patterns that broker submissions routinely obscure. What a company doesn't disclose is as informative as what it does.

Volume Without Degradation

DSI's automated Tier 1–2 approval — 75–85% of clean submissions — allows the same underwriting team to process dramatically higher volumes without degrading selection quality. Scale E&S volume without adverse selection risk.

The $100B E&S Opportunity

The US E&S market now exceeds $100B in premium and is growing faster than any other segment as admitted carriers shed complexity. MGAs are the primary underwriting vehicle for that growth. Speed and selection discipline are both essential.